

Towards Group-Conditional Coverage Guarantees via Group-Invariant Representations

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1. Introduction

Suppose that we have a deep neural network trained for a classification task $f : \mathcal{X} \rightarrow \mathcal{Y} = \{1, \dots, K\}$, $K \in \mathbb{N}^*$, in a supervised way on a finite set of labelled observations $\{(\mathbf{x}_i, y_i), i = 1, \dots, n\}$. Let $\mathfrak{X}$ be the representation of the data learned by the neural network, and p^y be the estimated probability of class y (for example, the calibrated softmax value of y). The split-conformal procedure (Papadopoulos et al., 2002; Vovk et al., 2022) is a powerful method to “transform” the class predictor f into a set predictor $\mathcal{C}_\alpha : \mathcal{X} \rightarrow 2^{\mathcal{Y}}$ where $2^{\mathcal{Y}}$ is the power set of $\mathcal{Y}$ with the valid *marginal* coverage guarantee:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}_\alpha(\mathbf{X}_{n+1})) \geq 1 - \alpha, \quad (1)$$

where $\alpha \in (0, 1)$ is a chosen tolerance to error. To do this, a non-conformity score function $\Delta : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ that quantifies the *strangeness* of any arbitrary couple $(\mathbf{x}, y)$ needs to be defined, often based on the estimated probabilities p^y output by the neural network (Johansson et al., 2017; Melki et al., 2024).

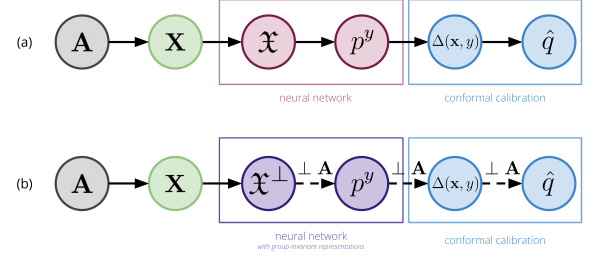
The conformal calibration step consists of estimating $\hat{q}$ as the $\frac{\lceil (m+1)(1-\alpha) \rceil}{m}$ quantile of the scores computed on a held-out calibration set of size m , and the prediction step creates, for a new example $\mathbf{x}_{n+1}$, the prediction set of classes $\mathcal{C}_\alpha(\mathbf{x}_{n+1}) = \{y : \Delta(\mathbf{x}_{n+1}, y) \leq \hat{q}\}$.

2. Group-Conditional Conformal Prediction

In practice, we may have at our disposal, in addition to the labelled observations $(\mathbf{x}, y)$, some auxiliary data $\mathbf{a} \in \mathcal{A}$ that provide additional information about the examples ($\mathcal{A}$ is considered discrete and finite). For instance, $\mathbf{a}$ may be a vector of variables describing the environmental conditions under which the example $\mathbf{x}$ was observed. We would thus like to provide the following *group-conditional* coverage guarantee:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}_\alpha(\mathbf{X}_{n+1}) \mid \mathbf{a}_{n+1} = a) \geq 1 - \alpha, \forall a \in \mathcal{A}. \quad (2)$$

The split-conformal prediction procedure described in the introduction is not enough to provide the group-conditional guarantee in Equation (2). This is mainly due to the influence of exogenous factors, some of which may be encoded in $\mathbf{a}$, that result in each group having a different $1 - \alpha$ quantile, $\hat{q}_{\mathbf{a}_1} \neq \dots \neq \hat{q}_{\mathbf{a}_t} \neq \hat{q}$ where $t = |\mathcal{A}|$ is the number of groups (panel (a) in the adjacent figure), thus requiring group-conditional calibration. Indeed, if all the conditional quantiles were equal to the marginal quantile $\hat{q}$, then the marginal procedure would be enough to guarantee group-conditional coverage. Therefore, the question we ask is: Can we obtain equalized conditional quantiles so that marginal calibration is enough to maintain conditional coverage?



3. Group-Invariant Representations

3.1. Sketch of the Method

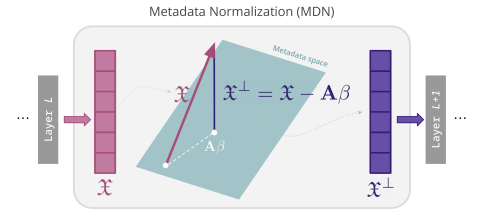
To answer the previous question, we propose an intervention that consists of breaking the chain of influence of $\mathbf{A}$ on the rest of the calibration procedure at the level of model training by making the model learn representations $\mathbf{X}^\perp$ that are *invariant to the auxiliary information* $\mathbf{A}$, instead of $\mathbf{X}$, as shown in panel (b) in the figure above.

What we wish to achieve is the invariance of the nonconformity scores (that we will simply denote as Δ) with respect to the auxiliary information, conditional on the learned representations. In other words, we want to learn $\mathbf{X}^\perp$ such that $\Delta \perp \mathbf{A} | \mathbf{X}^\perp$. If that is achieved, then we would have $P(\Delta | \mathbf{X}^\perp, \mathbf{A}) = P(\Delta | \mathbf{X}^\perp)$.

3.2. Learning with Metadata Normalization

Metadata Normalization (MDN), introduced by Lu et al. (2021), is a method that aims at learning model representations that are orthogonal to some auxiliary data. MDN is a simple layer that can be integrated into any neural network, at different stages, in order to “filter out” the impact of exogenous variables.

Consider the figure to the right showing a conceptual representation of a MDN layer inserted between a layer L and a layer $L+1$ of the neural network. The MDN layer essentially tries to model the representation $\mathbf{X}$ as a linear function of the auxiliary data by learning the linear model $\mathbf{X} = \mathbf{A}\beta + \mathbf{r}$, where $\mathbf{r}$ is the residual term and β are linear parameters to be estimated. This model basically tries to “separate” $\mathbf{X}$ into two parts: $\mathbf{A}\beta$, the component of $\mathbf{X}$ that is linearly related to the metadata $\mathbf{A}$, and the residual term $\mathbf{r} = \mathbf{X}^\perp$, which is orthogonal to the linear subspace spanned by the metadata. By integrating multiple MDN layers in the network’s architecture we would effectively disentangle the nonconformity scores from the auxiliary data. Preliminary results on the agricultural task of crop and weed image classification with environmental auxiliary data and MDN integration in the final layers of a neural network will be shown during the poster session.



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